Stroke Risk Prediction Model

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Background

- Strokes are caused by a blockage or reduction of blood flow to the brain
- Strokes are the 4th leading cause of death and disability in the US
- 80% of strokes are preventable

[1] stroke.org, *American Stroke Association*

Problem

Stroke risk can be assessed using multiple factors and symptoms. Risk calculators use these variables to estimate the likelihood of a stroke, but different prediction models assign different weights to each factor, leading to variations in risk estimates.

Stroke Prediction Model

Purpose

Use supervised machine learning to determine the weight of multiple symptoms and demographics towards the risk of stroke and predict the likelihood of stroke.

Dataset

Demographics: gender, age, stroke risk percentage

Symptoms: status of chest pain, high blood pressure, irregular heartbeat, shortness of breath, fatigue, dizziness, excessive sweating, persistent cough, vomiting, chest discomfort, cold hands/feet, sleep apnea, anxiety

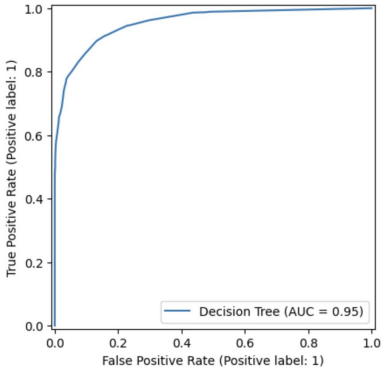
Methodology

1. Train several models using the stroke dataset. Dataset contains whether the patient had a stroke or not, allowing for validation of the model.

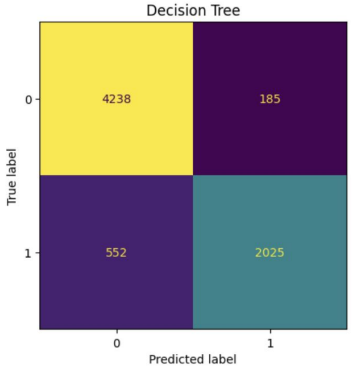
2. Compare models based on ROC-AUC, F1, and accuracy. ROC-AUC and confusion matrices are output.

3. Compare the weighting of coefficients in the models to determine which has the greatest impact on stroke risk.

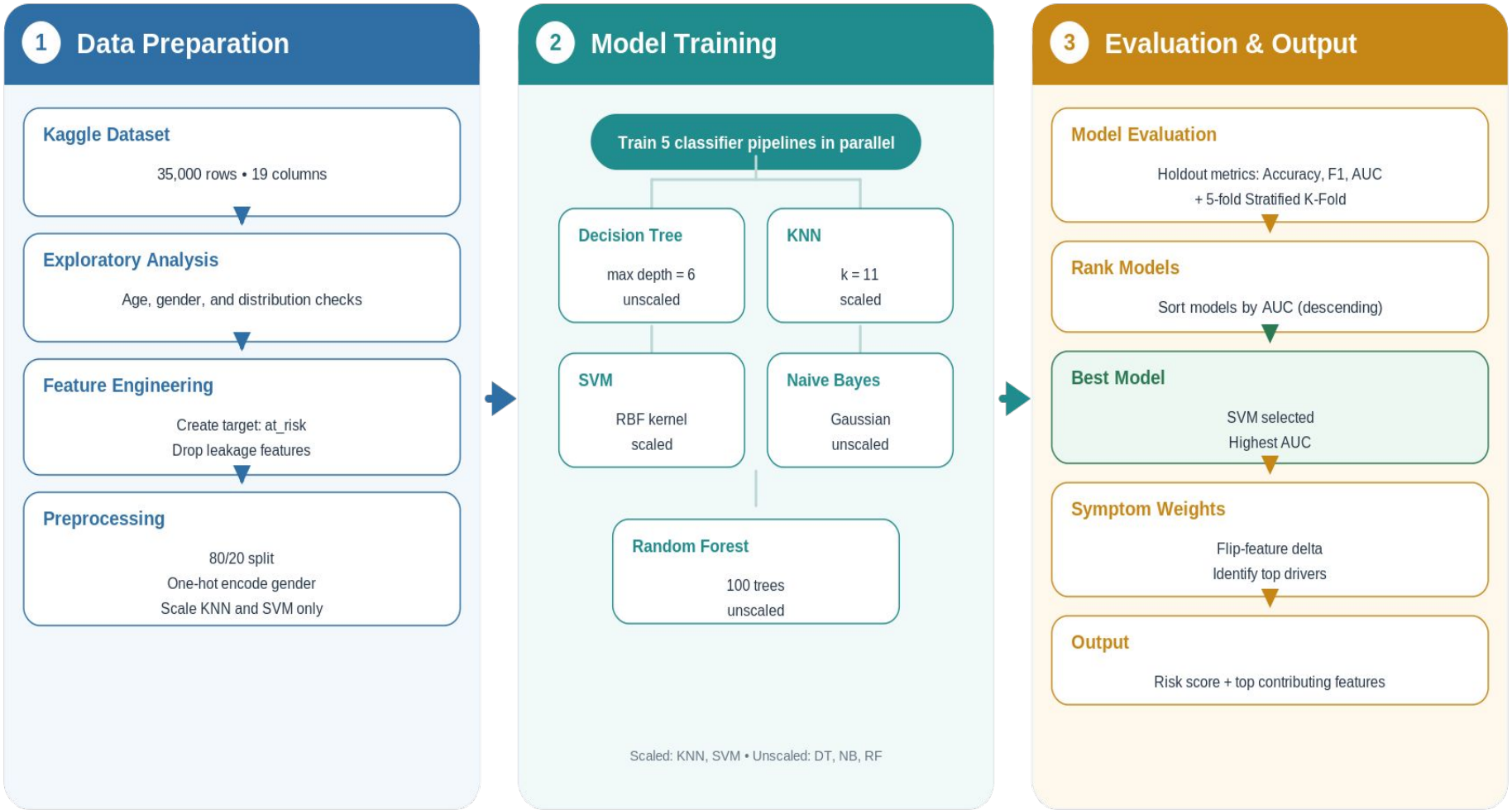
Receiver Operator Characteristics



Confusion Matrix



$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall}$$


1 Data Preparation
Kaggle Dataset
35,000 rows • 19 columns
Exploratory Analysis
Age, gender, and distribution checks
Feature Engineering
Create target: at_risk
Drop leakage features
Preprocessing
80/20 split
One-hot encode gender
Scale KNN and SVM only
2 Model Training
Train 5 classifier pipelines in parallel
Decision Tree
max depth = 6
unscaled
KNN
k = 11
scaled
SVM
RBF kernel
scaled
Naive Bayes
Gaussian
unscaled
Random Forest
100 trees
unscaled
Scaled: KNN, SVM • Unscaled: DT, NB, RF
3 Evaluation & Output
Model Evaluation
Holdout metrics: Accuracy, F1, AUC
+ 5-fold Stratified K-Fold
Rank Models
Sort models by AUC (descending)
Best Model
SVM selected
Highest AUC
Symptom Weights
Flip-feature delta
Identify top drivers
Output
Risk score + top contributing features

https://drive.google.com/file/d/11XA7qeKVXGbQGjIcLG1PYo8l5eZS0Ax2/view?usp=sharing

CONCLUSIONS

• *Model*

SVM was selected to be the optimal model for this dataset, suggesting the data is complex and has some nonlinear behavior. SVM excels with high dimensional datasets where kernels are used for high generalizability via maximum margins.

• *Coefficients*

Chest pain had the highest weight of 12.32% and high blood pressure following with 11.86%. These make sense as chest pain is a direct symptom of blockage or stress in the heart and high blood pressure is one of the primary factors associated with stroke.

THANK YOU

References

[1] American Stroke Association, “About Stroke,” *American Stroke Association*, 2024. https://www.stroke.org/en/about-stroke